

Average Rate of Change

ANSWER KEY



Computing average rate of change

- 1 Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2] = 2$

b $[1, 3] = 4$

c $[2, 5] = 7$

d $[-1, 1] = 0$

- 2 Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.

$$\frac{f(3) - f(0)}{3 - 0} = \frac{8 - 1}{3} = \frac{7}{3}.$$

- 3 Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

$$\frac{g(b) - g(a)}{b - a} = \frac{(3b - 5) - (3a - 5)}{b - a} = \frac{3(b - a)}{b - a} = 3. \text{ A linear function changes at a constant rate equal to its slope.}$$

Applications

- 4 A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

a Find the average rate of change of the height over $[0, 2]$. $\frac{h(2) - h(0)}{2 - 0} = \frac{20 - 0}{2} = 10$ m/s.

b Find the average rate of change of the height over $[1, 3]$. $h(1) = 15$ and $h(3) = 15$, so the average rate of change is $\frac{15 - 15}{2} = 0$ m/s.

c Interpret your answer to part b in the context of the ball's motion. Over $[1, 3]$ the ball rises and then falls back to the same height, so on average its height does not change.

- 5 The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.

$f(0) = 1$ and $f(4) = -8 + 8 + 1 = 1$, so the average rate of change over $[0, 4]$ is $\frac{1 - 1}{4} = 0$. The secant line through the endpoints is horizontal.

- 6 A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.

a Find the average rate of change of revenue over $[0, 4]$. $R(0) = 40$, $R(4) = 40 + 24 - 8 = 56$: rate $= \frac{56 - 40}{4} = 4$ thousand riyals/year.

b Find the average rate of change of revenue over $[4, 8]$. $R(8) = 40 + 48 - 32 = 56$: rate $= \frac{56 - 56}{4} = 0$ thousand riyals/year.

c Compare your two answers and interpret what they say about the company's growth. Revenue grew steadily during the first period but plateaued during the second — the average growth rate dropped to zero.

Comparing rates and estimating from a table

- 7 A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3] = \frac{17-5}{2} = 6$

b $[3, 6] = \frac{44-17}{3} = 9$

c $[6, 10] = \frac{92-44}{4} = 12$

- 8 Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g 's graph?

The average rate of change is increasing (6, 9, 12), suggesting g is concave up — consistent with g being a quadratic or otherwise increasingly steep function.

- 9 Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

$d_1: \frac{30-10}{4} = 5$ km/h. $d_2: \frac{36-4}{4} = 8$ km/h. The second traveler had the greater average speed over this interval.

Average rate of change and secant lines

- 10 Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

The average rate of change equals the slope of the secant line connecting the two points $(a, f(a))$ and $(b, f(b))$ on the graph of f .

- 11 For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.

$f(-1) = -1$, $f(2) = 8$. Rate $= \frac{8-(-1)}{2-(-1)} = \frac{9}{3} = 3$. Secant line: $y - 8 = 3(x - 2)$, i.e. $y = 3x + 2$.

Comparing average rate of change across different functions

- 12 Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

$f: \frac{16-1}{4} = 3.75$. $g: \frac{16-0}{4} = 4$. Over this interval, $g(x) = x^2$ had the slightly larger average rate of change.

- 13 Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

$f(4) = 16$, $f(8) = 256$. Rate $= \frac{256-16}{4} = 60$. This is much larger than the rate over $[0, 4]$ (3.75), illustrating that exponential functions grow at an accelerating average rate over successive equal-width intervals, unlike linear functions.

Average rate of change from a graph

- 14** Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.

From the graph: $f(0) = 3$, $f(4) = 3$. Average rate of change $= \frac{3-3}{4-0} = 0$, matching the fact that the parabola's vertex lies at the midpoint $x = 2$ and the two endpoints have equal height.

Units and interpretation in context

- 15** A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

$\frac{V(5) - V(0)}{5 - 0} = \frac{125 - 200}{5} = -15$ liters/minute. The negative sign means the tank is draining on average at 15 liters per minute over this interval.

- 16** A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.

$P(0) = 500$, $P(10) = 500e^1 \approx 1359$. Average rate of change $\approx \frac{1359 - 500}{10} \approx 86$ people/year.